

Problem-Solving Agents

Problem Formulation & State-Space Representation

Learning Objectives

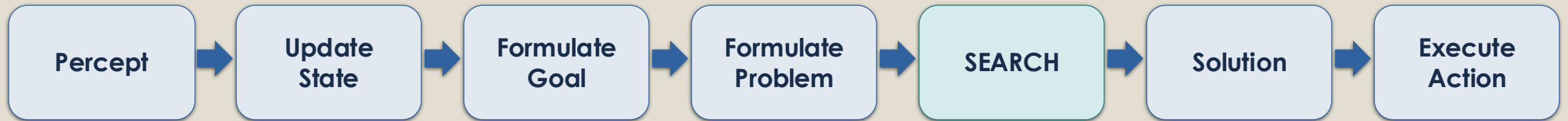
- Explain what a problem-solving agent does.
- Understand the five components of a well-defined problem.
- Formulate a route-finding problem using the Romania example.
- Explain state space, states, actions, successors, paths and solutions.
- Understand how problem formulation leads to search.

What is a Problem-Solving Agent?

- A problem-solving agent is a goal-based agent that finds a sequence of actions to reach a goal.
- The agent first formulates a goal and a problem.
- It then searches for a solution and executes the actions in that solution.

Goal → Problem Formulation → Search → Solution → Execute

Simple Problem-Solving Agent



The search function finds an action sequence that solves the problem.

Well-Defined Problem: Five Components

1. Initial State
Where the agent starts

2. Actions
What actions are possible

3. Transition Model
What state results from an action

4. Goal Test
Checks whether the goal is reached

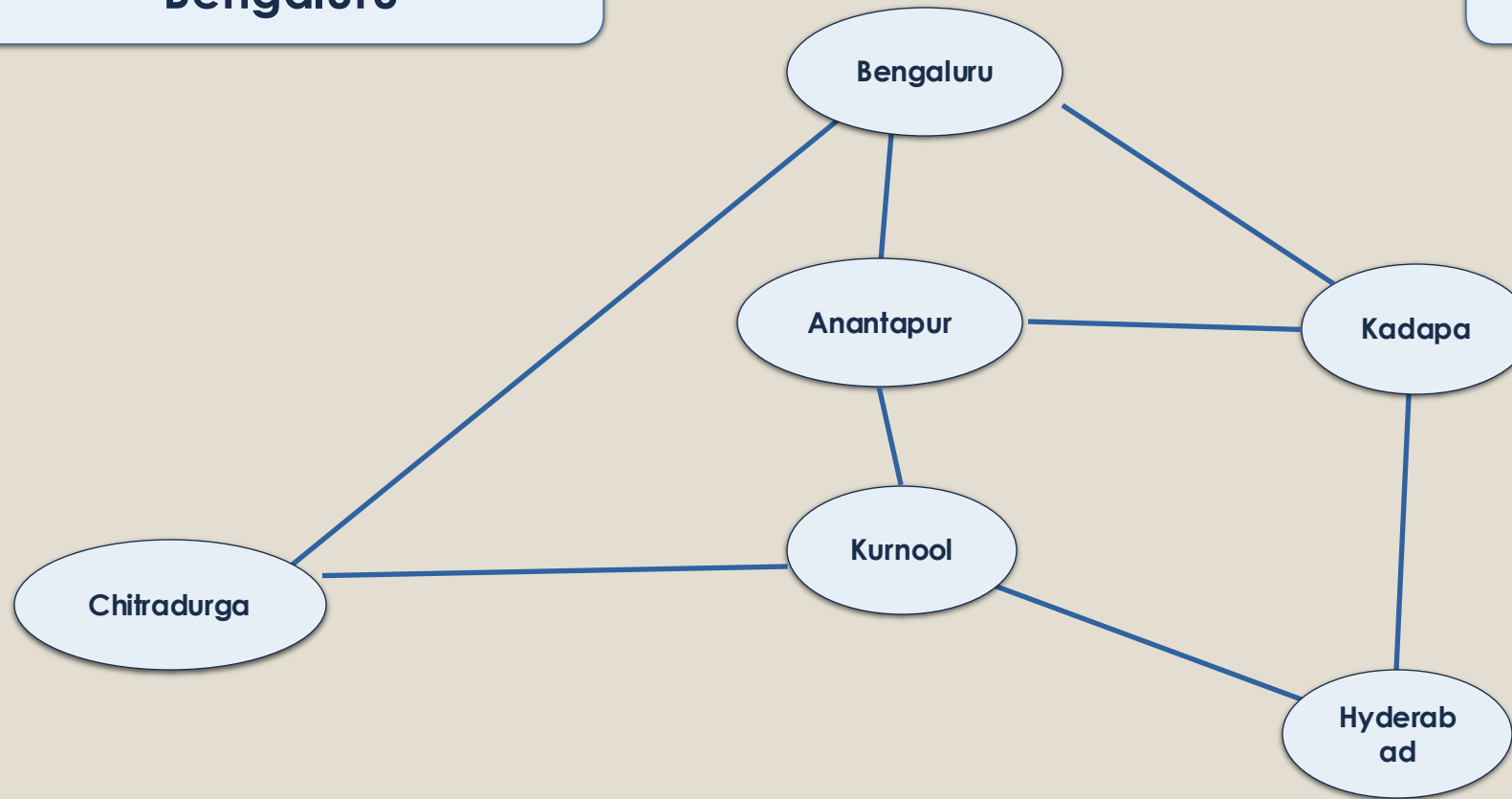
5. Path Cost
Numerical cost of a path

Example: Bengaluru → Hyderabad Route Finding

A simplified classroom state-space example using the textbook's problem-formulation framework

Initial State
Bengaluru

Goal
Hyderabad



Note: This is a simplified teaching graph, not a complete road network.

1. Initial State

In(Bengaluru)

- The initial state is the state in which the agent starts.
- In the Romania problem, the agent starts in Bengaluru.
- Notation used by the textbook: In(Bengaluru).

2. Actions

ACTIONS(In(Bengaluru))

Go(Anantapur)

Go(Chitradurga)

Go(KADAPA)

ACTIONS(s) returns the set of actions that can be executed in state s.

3. Transition Model

RESULT(s, a) → resulting state

**RESULT(In(Bengaluru),
Go(Anantapur))**

In(Anantapur)

Current State + Action → New State

4. Goal Test

Goal = In(Hyderabad)

Is current state In(Hyderabad)?

- YES → Goal reached
- NO → Continue searching

5. Path Cost

- A path-cost function assigns a numerical cost to each path.
- The cost should reflect the agent's performance measure.
- In route finding, the cost can be distance in kilometers.
- Other problems may use time, money, fuel, energy, or another measure.

Example: $140 \text{ km} + 99 \text{ km} + 211 \text{ km} = 450 \text{ km}$

State Space, Path and Solution

- State space = all states reachable from the initial state by any sequence of actions.
- State-space graph: nodes represent states; links represent actions.
- Path = sequence of states connected by a sequence of actions.
- Solution = a path that reaches a goal state.

Initial State + Actions + Transition Model → State Space

Complete Problem Formulation: Bengaluru → Hyderabad

Initial State

In(Bengaluru)

Actions

Go(Anantapur), Go(Chitradurga), Go(another connected city)

Transition Model

RESULT(s,a) gives the resulting state

Goal Test

Current state = In(Hyderabad)?

Path Cost

Total route distance / chosen cost measure

From Problem Formulation to Search



The next step is to study search strategies such as BFS, DFS, Uniform-Cost Search and A*.

Quick Revision

I	Initial State	Where do I start?
A	Actions	What can I do?
T	Transition Model	What happens after an action?
G	Goal Test	Have I reached the goal?
P	Path Cost	How expensive is this path?

Memory Trick: I – A – T – G – P

